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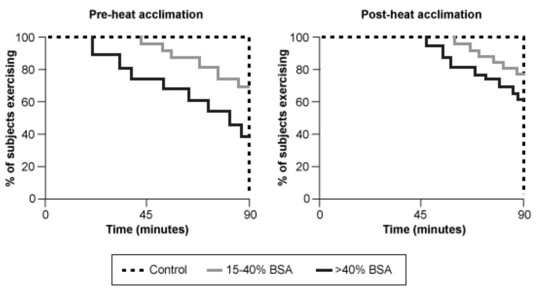


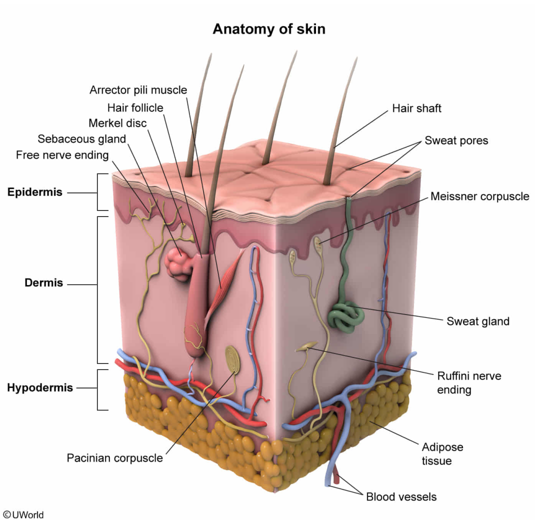
Figure 1 HA in controls and in skin graft recipients

Given that scar tissue forms when collagen-producing cells replace normal tissue after an injury, all the following processes would be impaired in the area of scarred skin that forms over a partial-thickness burn EXCEPT:

- A. resistance to ultraviolet radiation. [25%]
- ✔ B. insulation. [46%]
- C. hair growth. [12%]
- D. salt excretion. [16%]

Correct
46% answered correctly

Explanation:



Normal tissue that has been damaged is replaced by scar tissue, which is made primarily of collagen-producing cells (eg, fibroblasts). When scar tissue replaces the dermis and epidermis, the following normal skin functions are affected:

- **Protection from ultraviolet (UV) radiation**: Melanin, a dark pigment produced by epidermal melanocytes, absorbs and prevents UV light from damaging the DNA of other cells in the body. Melanin is transferred from the melanocytes to the keratinocytes, cells that produce the structural protein keratin. The lack of melanocytes (and consequently the lack of melanin) in the scar tissue would reduce the affected area's resistance to the harmful effects of UV radiation **(Choice A)**.

- **Hair growth**: Hair strands are composed of dead keratinized cells and are produced in the hair follicles, structures that contain both epidermal and dermal components. Therefore, scarring that replaces the epidermis and dermis would impair hair growth **(Choice C)**. Fingernails, toenails, and calluses are also protective keratinized structures derived from the skin.

- **Sweating**: Sweat glands arise from the dermis and secrete sweat (a mixture of water, salts, trace nitrogenous waste, and antimicrobial proteins) through ducts in the epidermis. Because scar tissue does not contain sweat glands, the scarred area would be incapable of salt excretion **(Choice D)**.

The subcutaneous layer of skin (also called the hypodermis) is composed of adipose tissue that buffers rapid fluctuations in body temperature by acting as a thermal insulator. Based on the passage, the subcutaneous layer is *not* damaged in partial-thickness burns, so the **insulation function of skin would not be affected**.

Educational objective:

Skin has many functions: The subcutaneous layer is composed of adipose cells that insulate the body, epidermal melanocytes prevent UV radiation from damaging the DNA of cells, keratinized derivatives of skin help protect the body from external injury, and dermal sweat glands secrete sweat onto the skin surface to regulate body temperature.

Time spent:0 sec **QID:**400935

Subject: **System:**

Subject: Biology

System: Skin and Immune Systems

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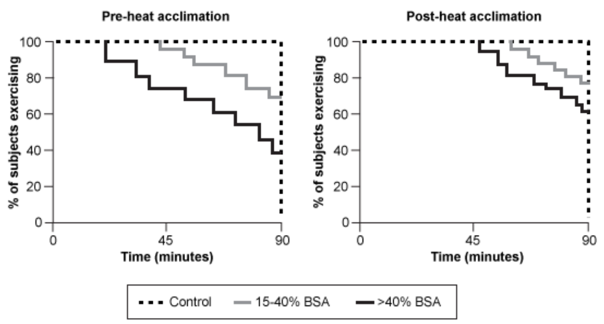


Figure 1 HA in controls and in skin graft recipients

Which of the following skin functions will most likely be impaired in tissue with superficial burns?

'

I. Keratinocyte maturation

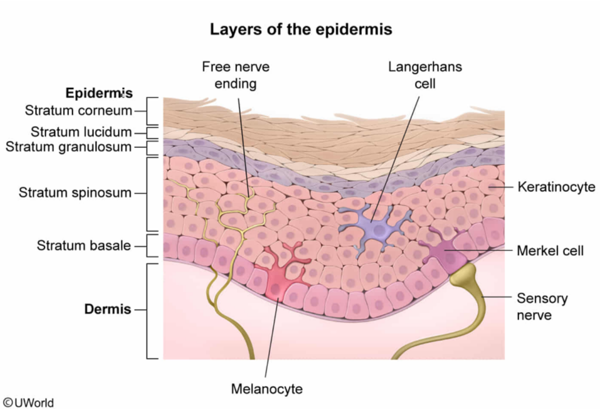
II. Immune surveillance

III. Shock absorption

- A. I and II only [47%]
- B. I and III only [22%]
- C. II and III only [15%]
- D. I, II, and III [14%]

Correct
47% answered correctly

Explanation:



The skin is divided into three main layers (ie, epidermis, dermis, and hypodermis). The epidermis can be further divided into five additional layers (strata) that contain Merkel cells, Langerhans cells, keratinocytes, and melanocytes. The **deepest layer of the epidermis**, or stratum basale, consists of a single **row of stem cells** that **continually divide** to give rise to new stem cells.

Of the two daughter cells produced from each mitotic division in the stratum basale, one cell remains in the basal layer (to continue proliferating) and the other begins differentiating into a **mature keratinocyte**. As constant cell division pushes the keratinocytes outward through the epidermal layers, they fill with keratin, flatten, and lose their organelles.

The outermost layer of the epidermis, the stratum corneum, is composed of 20–30 layers of these dead keratin-filled cells and functions as a physical barrier to protect against pathogens, ultraviolet light, water loss, and injury.

As stated in the passage, a **superficial burn damages only the epidermis**. Because the stratum basale, which is responsible for producing keratinocytes that undergo maturation, is part of the epidermis, **keratinocyte maturation will be impaired** in tissue with superficial burns **(Number I)**.

Epidermal Langerhans cells are immune (dendritic) cells that recognize and ingest antigens before migrating to nearby lymph nodes to present these antigens to T cells, activating the *adaptive immune response*. The adaptive immune response is highly targeted to eliminate a specific pathogen whereas the innate immune response is generalized to attack any foreign substance. Because Langerhans cells are part of the epidermis, **immune surveillance will be impaired** in tissue with superficial burns **(Number II)**.

The dermis (middle layer) is composed of connective tissue and contains blood vessels, sensory receptors, hair follicles, and sweat (sudoriferous) and oil (sebaceous) glands. The hypodermis, or the subcutaneous layer beneath the dermis, is composed primarily of adipose tissue that protects the internal organs by acting as a shock absorber and insulator. Because superficial burns damage only the epidermis (not the hypodermis), **shock absorption will not be impaired** in tissue with superficial burns **(Number III)**.

Educational objective:
The skin is divided into the epidermis (outermost layer), dermis (middle layer), and hypodermis (innermost layer). The epidermis acts as a physical barrier that separates the organism from the external environment. The dermis contains blood vessels, immune cells, sensory receptors, sweat and oil glands, and hair follicles. The hypodermis is composed of insulating and shock-absorbing adipose tissue.

Time spent:0 sec	**QID**:400936
Subject: Biology	**System:** Skin and Immune Systems

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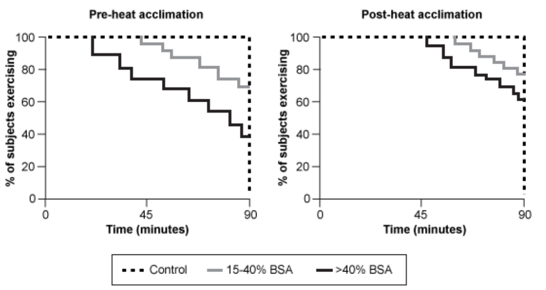


Figure 1 HA in controls and in skin graft recipients

Which table likely illustrates the change in average internal body temperature that occurred during heat tolerance testing pre-HA and post-HA in grafted and control subjects? Assume all participants had an average internal body temperature of 37°C before heat-tolerance testing began.

○ A.

Group	Condition	Δ Internal temperature (°C)
Control	Pre-HA Post-HA	1.8 1.8

Grafted (≥15% BSA)	Pre-HA Post-HA	2.1 2.1

[6%]

B.

Group	Condition	Δ Internal temperature (℃)
Control	Pre-HA Post-HA	2.3 2.2
Grafted (≥15% BSA)	Pre-HA Post-HA	1.7 1.7

[7%]

C.

Group	Condition	Δ Internal temperature (℃)
Control	Pre-HA Post-HA	2.0 2.5
Grafted (≥15% BSA)	Pre-HA Post-HA	1.6 2.1

[23%]

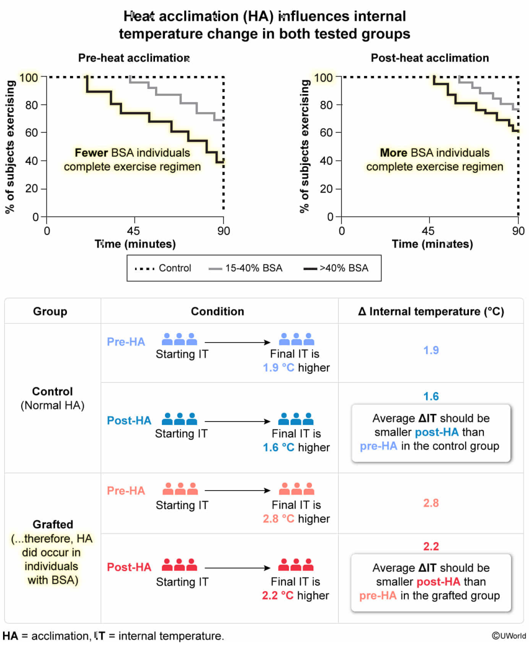
D.

Group	Condition	Δ Internal temperature (℃)
Control	Pre-HA Post-HA	1.9 1.6
Grafted (≥15% BSA)	Pre-HA Post-HA	2.8 2.2

[62%]

Correct
62% answered correctly

Explanation:



According to the passage, heat acclimation (HA), the process by which the body increases tolerance to environmental heat, can be induced by repeatedly exercising at elevated temperature and humidity. To determine if skin graft recipients could undergo HA, researchers performed testing on skin graft recipients (divided into two groups based on percentage of grafted BSA) and controls with no history of receiving grafts. The researchers measured the participants' heat tolerance *before* and *after* a seven-day period of HA training by determining their ability to complete a 90-min exercise regimen as well as the change in their internal temperature during the exercise. Heat tolerance testing was halted before 90 min if the internal temperature (IT) of subjects exceeded 40°C.

Pre-HA, all controls were able to complete the 90-min exercise regimen, indicating that controls were able to sustain ITs below 40°C throughout HA training, as shown in Figure 1. However, in both skin graft recipient groups, some members were unable to complete the regimen, meaning that these individuals' ITs rose above 40°C. Given that the question assumes that all participants had the same average IT prior to beginning exercise (37°C), this signifies that **graft recipients** were **less able to tolerate heat compared with controls** and had **a larger average increase in IT during exercise**.

Post-HA, all controls were able to complete the 90-min regimen, but the individuals with skin grafts showed improved performance as more were able to make it to the 90-min endpoint. Therefore, the average change in IT of the **grafted groups** was smaller post-HA than pre-HA. In addition, the passage states that HA occurs normally in healthy individuals. Therefore, **control individuals** would also be expected to show some increase in heat tolerance (a smaller post-HA change in IT than pre-HA change in IT). The table shown in **Choice D** is the only option that illustrates the expected **increase in heat tolerance** considering that **HA occurred in both groups**.

(Choices A and B) HA would affect the change in IT of *both* tested groups.

(Choice C) The post-HA change in IT would be expected to decrease (*not* increase) in both tested groups.

Educational objective:
The skin plays a key role in the thermoregulation of body temperature, and injury can impair its ability to tolerate significant temperature changes.

Time spent:0 sec
Subject:
Biology

QID:400937
System:
Skin and Immune Systems